AGENTIC AI KNOWLEDGE BASE • WEATHER & PLAY COMPREHENSIVE INDEX

Weather Parameters & Day/Night Game Matrix

1. Core Parameters & RAG Indexing Scope

This knowledge base governs recommendations based strictly on four primary meteorological indicators: **Apparent Temperature**, **Wind Speed**, **Is Day (Day/Night cycle)**, and **Precipitation**. Activities are strictly segregated into Day and Night operational windows across diverse indoor and outdoor disciplines.

Key Evaluated Parameters

- **Apparent Temperature (°C):** Human thermal perception accounting for wind chill and humidity.
- **Wind Speed (km/h):** Aerial friction affecting aerodynamics in sports.
- **Is Day (1 / 0):** Solar illumination flag (1 = Daylight, 0 = Night).
- **Precipitation (mm/h):** Liquid/solid water accumulation rate dictating outdoor feasibility.

2. Comprehensive Master Decision Matrix

Condition Profile	Apparent Temp	Wind Speed	Is Day	Precipitation	Primary Day Activities	Primary Night Activities
M1: Clear & Mild	18°C - 28°C	< 15 km/h	1 (Day)	0.0 mm	Football, Cricket, Tennis, Golf, Athletics, Kayaking	Night Cricket, Night Jogging, Outdoor Basketball
M2: Clear & Cold	-5°C - 8°C	< 20 km/h	1 (Day)	> 0 (Snow)	Skiing, Snowboarding, Sledding, Winter Hiking	Indoor Hearth Games, Night Ice Skating Rink
M3: Hot & Humid	> 32°C	< 20 km/h	1 (Day)	0.0 mm	Water Polo, Kayaking, Swimming, Shaded Golf	Night Swimming, Indoor Bowling, Foosball
M4: Rainy & Stormy	10°C - 22°C	> 25 km/h	1 or 0	> 3.0 mm	Indoor Swimming, Billiards, Carrom, Table Tennis	Carrom, Table Tennis, Chess, Foosball, Darts
M5: Clear Calm Night	12°C - 20°C	< 10 km/h	0 (Night)	0.0 mm	N/A (Daytime profile)	Night Cycling, Stargazing Walks, Night Tennis
M6: High Wind Day	15°C - 25°C	> 35 km/h	1 (Day)	0.0 mm	Paragliding (ridge soaring), Kite Flying, Indoor Squash	Indoor Bowling, Billiards, Arcade Gaming

3. Enriched Game Catalog by Weather Condition & Time Cycle

Profile A: Optimal Outdoor Conditions (Clear, Low Wind, Zero Precipitation)

APPARENT TEMP	WIND SPEED	PRECIPITATION	CYCLE SPLIT
15°C - 28°C	**0 - 15 km/h**	**0.0 mm**	**Day vs Night**

Outdoor Games (Day & Night)

- **Daytime Outdoor:** Football/Soccer, Cricket, Hockey (turf), Tennis, Golf, Baseball, Athletics, Skateboard, Trail Running, Kayaking, Paragliding.
- **Nighttime Outdoor:** Floodlit Football, Night Cricket, Floodlit Tennis, Jogging, Outdoor Basketball, Night Cycling.

Indoor Games (Day & Night)

- **Daytime / Nighttime Indoor:** Carrom, Table Tennis, Bowling, Foosball, Billiards/Pool, Snooker, Squash, Badminton, Indoor Basketball, Darts, Chess.

Profile B: Adverse Precipitation & Storm Conditions (High Rain/Snow)

APPARENT TEMP	WIND SPEED	PRECIPITATION	CYCLE SPLIT
Any Range	**> 20 km/h**	**> 2.5 mm/h**	**Strict Indoor**

Outdoor Games (Restricted)

- **Daytime Outdoor:** Mud Football, Extreme Kayaking (experienced only), Snow Skiing/Snowboarding (if sub-zero snow).
- **Nighttime Outdoor:** Generally suspended due to safety hazards, poor visibility, and slippery surfaces.

Indoor Games (Day & Night)

- **Daytime Indoor:** Indoor Swimming, Squash, Badminton, Rock Climbing (indoor gym), Gymnastics, Bowling, Foosball, Table Tennis, Carrom, Billiards.
- **Nighttime Indoor:** Board Games, Chess, Darts, Arcade Gaming, Billiards, Carrom, Foosball, Table Tennis tournaments.

Profile C: High Wind & Thermal Stress Conditions (Extreme Heat or High Wind)

APPARENT TEMP	WIND SPEED	PRECIPITATION	CYCLE SPLIT
> 32°C or < 0°C	**> 30 km/h**	**0.0 mm**	**Modified Play**

Outdoor Games (Specialized)

- **Daytime Outdoor:** Paragliding, Kite Flying, Water Polo, Kayaking (heat); Ice Hockey, Skiing, Sledding (sub-zero).
- **Nighttime Outdoor:** Cooler night running, Night Golf, Illuminated Skating.

Indoor Games (Climate Controlled)

- **Daytime / Nighttime Indoor:** Air-conditioned Table Tennis, Carrom, Foosball, Bowling, Billiards, Darts, Chess, Indoor Basketball.